ASSIGNMENT 2: RESEARCH-BASED IMPLEMENTATION STRATEGY
Course: ISO/IEC 42001 Lead Implementer

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Case Study: Amazon AI Recruiting Tool (2014-2018)
Date: 16 October 2026

STEP 1: CASE STUDY SELECTION

Selected: Amazon Secret AI Recruiting Tool - The system learned to penalize resumes containing"women's" after training on 10 years of male-dominated resumes. Scrapped 2018. Source: Reuters 2018.

SECTION 1: EXECUTIVE SUMMARY & CONTEXT

1.1 Overview
Amazon wanted efficiency: thousands of CVs/days for tech roles. Built ML model to rank CVs 1-5 stars based on historical hiring data.
Result: Model replicated past bias. Downgraded all-women's colleges.
Attempt to neutralize "women's" failed - model found proxy variables.
Project closed 2017.

1.2 Context of Organization - Clause 4.1
Internal Issues:
- Scale pressure, automate-everything culture
- No AI governance function, no AI policy
- No documented risk appetite
External Issues:
- #MeToo movement 2017, GDPR 2018 coming
- Title VII, EEOC guidance on AI hiring
- Public expectation of fair hiring from Big Tech

Failure: No Clause 4.1 analysis done, high-risk context not identified.

1.3 Interested Parties - Clause 4.2

TABLE 1: INTERESTED PARTIES ANALYSIS

No	Interested Party	Need / Requirement	Impact of Failure
1	Female Applicants	Fair, Non-discriminatory screening	Exclusion, economic loss
2	HR & Recruiters	Tool that augments fair judgment	Loss of trust
3	Regulators (EEOC)	Compliance with Title VII	Investigation, fines
4	Shareholders / Board	Protect brand & ESG value	Reputational damage
5	Media / Civil Society	Transparency	Global negative case study
6	Women's Colleges	Equal recognition	Partnership damage

SECTION 2: ROOT CAUSE ANALYSIS THROUGH ISO/IEC 42001

This was management system failure, not code bug.

TABLE 2: ROOT CAUSE MAPPING TO ISO 42001 CLAUSES

Failure Observed	ISO 42001 Clause Violated	What Clause Requires	How it Caused Technical Failure
No bias risk considered	6.1.2 AI Risk Assessment	Identify & assess AI risks	Team never looked for bias
No impact on gender considered	6.1.4 AI System Impact Assessment	Assess impacts on individuals	Societal harm ignored
10-year CVs with no datasheet	7.4 / 7.5 Data for AI Systems	Data provenance & quality	Biased data entered model
Team only ML engineers	7.2 Competence	Competence in Responsible AI	No ethicist to flag issue
Tested only accuracy, not fairness	8.2 / 8.4 Design, Verification	Define fairness acceptance criteria	Biased model passed gate
Bias found 2015, no formal action	10 Improvement / 9.3 Reporting	Nonconformity & corrective action	Ad-hoc patch, not systematic

Key Lesson: Without Clause 6, you don't see risk. Without Clause 7, biased data enters. Without Clause 8, biased model ships.

SECTION 3: AI RISK AND IMPACT ASSESSMENT

Scenario: I am Lead Implementer hired BEFORE deployment in 2014.

3.1 Recreated Impact Assessment - Clause 6.1.4

System: Automated CV Screening & Ranking
Intended Use: Rank top 5 for recruiter review
Lifecycle: Design & Development

TABLE 3: ADVERSE IMPACT ASSESSMENT (CLAUSE 6.1.4)

Impact Type	Description	Severity	Likelihood	Affected Party
Societal / Ethical	Amplifies gender gap in tech, penalizes ''women's	Very High	High	Society / Women
Fundamental Rights	Violates right to non-discrimination Action	Very High	High	Female applicants
Legal / Compliance	Breaches Title VII, EEOC disparate impact	High	Medium	Amazone Legal
Reputational	Public exposure as discriminatory employer	High	High	Amazone Brand
Operational	Loss of diverse talent, overfit to past	Medium	High	Hiring Managers

Three Specific Impacts Ignored in Reality:

1. Societal: Reinforces stereotype tech = male
2. Ethical: Two identical CVs get different scores due to proxy
3. Individual: Candidate never knows why rejected, no recourse

3.2 Statement of Applicability - 5 Mandatory Controls

TABLE 4: STATEMENT OF APPLICABILITY (SoA) - SELECTED CONTROLS

Control ID	Control Title	Why Mandatory	How it prevents Recurrence
A.7.2	Data for AI Systems	Prevents training on undocumented biased data	Requires datasheet, gender split, balancing
A.7.4	Bias Assessment & Treatment	Directly addresses penalizing women's	Mandates 4/5ths rule, statistical parity – fails if biased
A.8.2	Transparency & Explainability	Black-box hid that ''women's'' was negative	Requires SHAP/LIME – negative feature visible early
A.6.2.4	AI System Impact Assessment procedure	Forces board-level review BEFORE build	Gate: No HR/Legal/Ethics approval = no dev
A.9.3	Reporting of concerns	2015 bias found but no channel to escalate	Safe whistleblowing to AI Ethics Borad, triggers Clause 10

Plus: A.5.3 AI Objectives - define fairness as objective, not just accuracy.

SECTION 4: 6-MONTH IMPLEMENTATION ROADMAP - RESCUE PLAN PDCA Model

TABLE 5: 6-MONTH ROADMAP

Phase	Month	Clause / Control	Action	Deliverable
PLAN	Month 1	Clause 5, 4.1, 4.2	Form AI Governance Board: CTO, CHRO, L EGAL, Diversity, external ethicist. Draft AI Policy A.2	Signed AI Policy, Context Report
PLAN	Month 2	Clause 6.1.2 ,6.1.4	Risk & Impact Assessment. Classify as HIGH-RISK AI. Halt deployment	Risk Register, Impact Assessment
DO	Month 3	Clause 7.2, 7.4, 7.5	Data remediation: datasheets, balance dataset, Responsible AI training	Clean dataset, Training records
DO	Month 4	Clause 8, A.7, A.8	Re-design: adversarial debiasing, Model Cards, human-in-loop, SoA	Model Card SoA Document
CHECK	Month 5	Clause 9.1, 9.2, 9.3	Red-team audit with 100 identical CVs, internal audit, Management Review	Audit Report, MR Minutes
ACT	Month 6	Clause 10, A.8.4	Decision: Retire ranking model-skills-matching only. Publish transparency report	Transparency Report, Lessons Learned

Outcome: Transform from ad-hoc ML project to trustworthy AIMS.

SECTION 5: CONCLUSION AND REFLECTION

5.1 Limitations of ISO/IEC 42001:

1. Management System, Not Product Fix: Clauses 4-10 manage HOW you govern, not IF business model is viable. Cannot make biased hiring strategy profitable.
2. Risk-Based Control Selection: Annex A has 38 controls A.2-A.10. Under Clause 6.1.3, org chooses controls in SoA. Weak culture can exclude A.5 or A.6.
3. Needs Leadership Commitment: Clause 5 and A.3.2 require top management. If culture rewards speed, policy A.2 is just paper.
4. Cannot Fix Data Reality: A.7 Data (A.7.5 provenance) documents bias but does not solve societal lack of women in tech pipeline.

5.2 What Standard Cannot Fix:

- Flawed Business Model: Clause 6.1.4 assesses impact on society, not financial viability. Will standard fix flawed model? No.
- Culture & Incentives: Even with A.3.3 reporting and A.9.2 responsible use, if bonuses tied to speed, no one reports.
- Reputation Post-Failure: Clause 10 and A.8.4 incident communication is reactive. Certificate does not restore trust.
- Technical Limits: A.6 Lifecycle (A.6.2.6 monitoring, A.6.2.8 event logs) requires monitoring but cannot prevent hallucination that science cannot explain.

5.3 Conclusion:

ISO 42001 would have prevented public failure via risk assessment, data governance, validation gate. But it would not solve deeper issue: Amazon needed to hire more women first to have balanced data. Standard is necessary but not sufficient. Needs genuine ethical culture.

REFERENCES

- ISO/IEC 42001:2023 Clauses 4.1, 4.2, 5, 6.1.2, 6.1.4, 7.2, 7.5, 8.2, 8.4, 9.1, 9.3, 10. Annex A Controls A.2, A.6.2.4, A.7.2, A.7.4, A.8.2, A.9.3, A.8.4, A.3.2, A.3.3, A.9.2
- Reuters: Dastin, J. (2018). Amazon scraps secret AI recruiting tool that showed bias against women.
- Gebru et al. (2021). Datasheets for Datasets.
- NIST AI RMF 1.0